Research Documentation

Online product reviews contain valuable information about customer satisfaction; however, the sheer amount of product reviews makes manual analysis inefficient. Sentiment analysis is a Natural Language Processing technique used to identify, extract, and quantify the emotional tone and subjective data found in text (Bellar et al., 2024). This project develops a sentiment classification system that will automatically categorize customer reviews into negative, neutral, and positive sentiment categories using Natural Language Processing and Transformer-based models. This can help companies determine overall customer sentiment trends in real time, detect recurring complaints about items, and highlight the strengths of their products (Ghatora et al., 2024).

Before the modernization of deep learning, analysts relied on rule-based methods, the bag of words, and term frequency inverse document frequency. Rule-based methods used preassigned sentiment scores for words, which would struggle to find sarcasm, as well as being unable to use context clues to detect what the user was actually saying (Gupta et al. 2024). Bag of words represented text as a collection of word counts, not paying attention to grammar or the order of the words, which caused it to fall short in regard to semantic meanings, thus allowing for limited accuracy. Term frequency inverse document frequency weighted words based on their frequency relative to how common they are across the whole data set but still struggled with contextual relationships between words (Nandwani & Verma, 2021). These traditional methods missed the nuances of sentence structure and context, which is where Transformer-based models come into play.

In a comprehensive study comparing BERT, LSTM, SVM, and lexicon-based models, BERT-based models demonstrated superior performance across all primary metrics achieving an accuracy of 86.51% and an F1-Score of 0.8653 on standardized sentiment datasets. This represents a pertinent performance advantage over traditional methods like Support Vector Machines and lexicon heavy approaches which typically have lower accuracies of 79.12% and 78.45%, respectively (Wang, 2024). While BERT does require a higher computational cost as it takes longer to train, the increase in the predictive precision is critical for the complex context of customer reviews. In addition, research emphasizes that this approach is necessary because sentiment analysis is not limited to binary outcomes. Using a three-class sentiment scale (negative, neutral, and positive) allows for a more accurate representation of customer opinions.

BERT accomplishes this through its self-attention mechanism, which enables the model to examine every word in a sentence simultaneously and determine the importance of surrounding words within their context (Nandwani & Verma, 2021). An example would be the ability to distinguish between “this product is bad” and “this product is not bad”, since it can understand how “not” interacts with “bad”(Nandwani & Verma, 2021). Using pretrained weights from large datasets, models like BERT allow for a contextual understanding of language. The extreme usefulness of BERT comes from transfer learning. Rather than having to start from scratch, the pretrained BERT weights will be fine-tuned on our Amazon review dataset. Each review has a rating from 1 to 5 stars, which will be the original indicator of customer satisfaction. The numerical ratings are split up into 3 different categories: 1 to 2 stars will be negative, 3 stars will be neutral, and 4-5 stars will be positive. This will allow the transformer model to learn the relationship between customer language and the overall sentiment. This is especially useful for product reviews because customers use complex language, mixed opinions, and contextual phrases that the traditional sentiment methods may incorrectly classify. By adding the classification layer on top of the pretrained model and training it on our labeled data, the model adapts its linguistic knowledge to become a highly accurate sentiment classifier. This is essential because by using this pretrained model, there is no need for the large volume of computing resources that would be required to train a brand-new language model from the ground up. Fine-tuning allows the BERT model to keep the general understanding of the language while adjusting its weights to better identify sentiment patterns.

References

Bellar, O., Baina, A., & Ballafkih, M. (2024). *Sentiment analysis: Predicting product reviews for e-commerce recommendations using deep learning and transformers*. *Mathematics, 12*(15), 2403. https://doi.org/10.3390/math12152403

Ghatora, P. S., Hosseini, S. E., Pervez, S., Iqbal, M. J., & Shaukat, N. (2024). *Sentiment analysis of product reviews using machine learning and pre-trained LLM*. *Big Data and Cognitive Computing, 8*(12), 199. https://doi.org/10.3390/bdcc8120199

Gupta, S., Ranjan, R., & Singh, S. N. (2024). *Comprehensive study on sentiment analysis: From rule-based to modern LLM-based systems*. *arXiv*. https://doi.org/10.48550/arXiv.2409.09989

Nandwani, P., & Verma, R. (2021). *A review on sentiment analysis and emotion detection from text*. *Social Network Analysis and Mining, 11*(1). https://doi.org/10.1007/s13278-021-00776-6

Wang, F. (2024). *Comparative evaluation of sentiment analysis methods: From traditional techniques to advanced deep learning models*. *Proceedings of the CONF-MLA 2024 Workshop: Neural Computing and Applications*, 23–29.